SkillUp

Reading: Generating a BPMN Process Model from an Email Thread or Meeting Summary

Estimated time: 7 minutes

Introduction

Business processes are often embedded within the daily exchanges in emails and meeting summaries. The main challenge is turning these informal, unstructured communications into structured process models, such as Business Process Model and Notation (BPMN) diagrams used for process analysis, automation, and improvement. In this reading, we'll explore actionable methods for extracting process information from unstructured sources and discuss the steps for converting this information into BPMN models using generative AI.

Identifying process elements in unstructured communication

Unstructured communication, such as meeting notes or an email thread, often hides process steps instead of presenting them in a clear order. Your job is to spot these elements and connect them to a BPMN model.

Focus on three key pieces:

- **Activities:** These are core actions or tasks. For example, "After receiving the client's request, I will update the tracking system" includes the activities "receive client request" and "update tracking system."
- **Decision points:** These are moments where choices or conditions are made. For instance, "If the request is urgent, escalate to the manager."
- **Responsible parties:** The roles performing the tasks, such as sales representative, project manager, or client.

Watch for trigger words such as "when," "if," or "after," since they often signal sequence or decision logic. Also, notice the implied responsibilities. For example, "Let me know when you've sent the document" suggests a handoff to the next person.

Structuring prompts for AI-driven extraction

You need clear, structured prompts to move from identifying process elements to extracting them with AI. A good prompt tells the AI exactly what to find and how to present the results. You can follow these steps:

1. State the elements you want extracted.
   i. Example prompt: "Extract all activities, decision points, and responsible parties from the following meeting summary. Present the results as a list of steps, each with an assigned role and any related conditions."
2. Filter out irrelevant details.
   i. Example prompt: "Summarize only the process steps in this email chain, ignoring greetings, status updates, and unrelated topics. Show the results as a sequential list suitable for process modeling."
3. Apply the prompt to honest communication.
   i. Meeting excerpt: "After the client submits the application, the sales representative reviews the details. If any information is missing, they contact the client for clarification."
      a. From this excerpt, the AI should return activities (submit application, review details, contact client), a decision point (if information is missing), and responsible parties (client, sales representative).

By tailoring your prompt, you help the AI focus on actionable content, making the output more relevant for BPMN modeling.

Recognizing system architecture requirements within technical discussions

Technical discussions often contain hidden process requirements within broader topics such as data flows, integrations, and system interactions. For example, the statement "The system should automatically send a notification to the client once payment is received" includes both an activity (send notification) and a system integration point (payment receipt triggers notification).

To guide AI in extracting these elements, you can use a prompt such as:

"Identify system interactions, data exchanges, or integrations in the following project notes. For each, specify the trigger, action, and involved systems."

For example: "When a new user registers, their info is sent to the CRM, and a welcome email is triggered."

- Trigger: A new user registers (a task the user performs).
- System interaction: The registration system sends the user's information to the CRM (a data exchange between two systems).
- Activity: The system sends a welcome email (an automated action performed by the system).

This approach ensures that system touchpoints are captured for BPMN modeling. Having recognized these technical elements, let's move on to the practical conversion of extracted process information into a BPMN model.

Converting extracted elements into a BPMN model

Once you have extracted activities, decisions, roles, and system interactions, the next step is to map them to BPMN elements:

- Activities are mapped to Tasks
  - For example: "Sales rep reviews request" becomes a Task.
- Decision points are mapped to Gateways
  - For example: "If urgent, escalate" becomes an Exclusive Gateway.
- Responsible parties are represented by Lanes or Pools
  - Example: "Manager approves" is placed in the Manager's lane.
- System interactions are represented by Message flows or Service tasks
  - Example: "System sends notification" becomes a Service Task.

Before creating the model, review the extracted steps for completeness and logical flow. Look for missing handoffs, unclear decision logic, or ambiguous responsibilities. For instance, if approval is required but not mentioned, confirm this with the process owner or check the source material.

With your elements clearly mapped, you are ready to refine the AI-generated model and ensure it accurately and clearly reflects the process.

Refining AI-generated BPMN models through iterative prompt engineering

Initial outputs from generative AI might be incomplete or inaccurate, especially when the source material is complex or ambiguous. You can improve results by refining your prompts step by step. Follow these steps:

1. **Review** the initial BPMN elements generated from your prompt.
2. **Identify** gaps, ambiguities, or sequence errors in the output.
3. **Refine** your prompt with more detail or context.
   i. For example, "In your previous output, you missed the manager's approval step. Please ensure the new process model includes all decision points and approvals."
4. **Repeat** the process until the extracted steps accurately reflect the real-world process.

For example:

1. Review: You check the AI's BPMN output.
2. Identify: You notice the escalation step is missing.
3. Refine: You revise the prompt to say, "Include all escalation scenarios, such as what happens if a request is urgent."
4. Repeat: You run the new prompt, confirm that the escalation step is included, and continue refining until the model accurately matches the process.

By steadily refining prompts and adding business context, you ensure the resulting BPMN model is accurate, actionable, and ready for process improvement or automation.

Conclusion

Transforming unstructured communications such as emails and meeting notes into BPMN process models requires carefully identifying activities, decisions, roles, and system interactions. Using well-structured prompts and refining AI outputs through iterative feedback, you can extract relevant process details and map them to BPMN elements. This approach ensures your process models are accurate, actionable, and valuable for process improvement and automation initiatives.

Summary

In this reading, you learned that:

- Identify activities, decision points, and responsible parties in unstructured notes using trigger words so you can map them accurately to a BPMN model.
- Use precise prompts that specify elements to extract, filter noise, and request structured outputs to produce actionable, BPMN-ready results.
- Extract triggers, system interactions, and activities from technical discussions to capture integrations and data flows needed for BPMN modeling.
- Map activities to tasks, decisions to gateways, roles to lanes or pools, and system interactions to message flows or service tasks, then review for completeness and logic.
- Iteratively review outputs, identify gaps, refine prompts with added context, and repeat until the BPMN model accurately reflects the real process.